



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-12

Experiment 12. Minimax Algorithm

Aim

To implement the Minimax algorithm for gaming using Python.

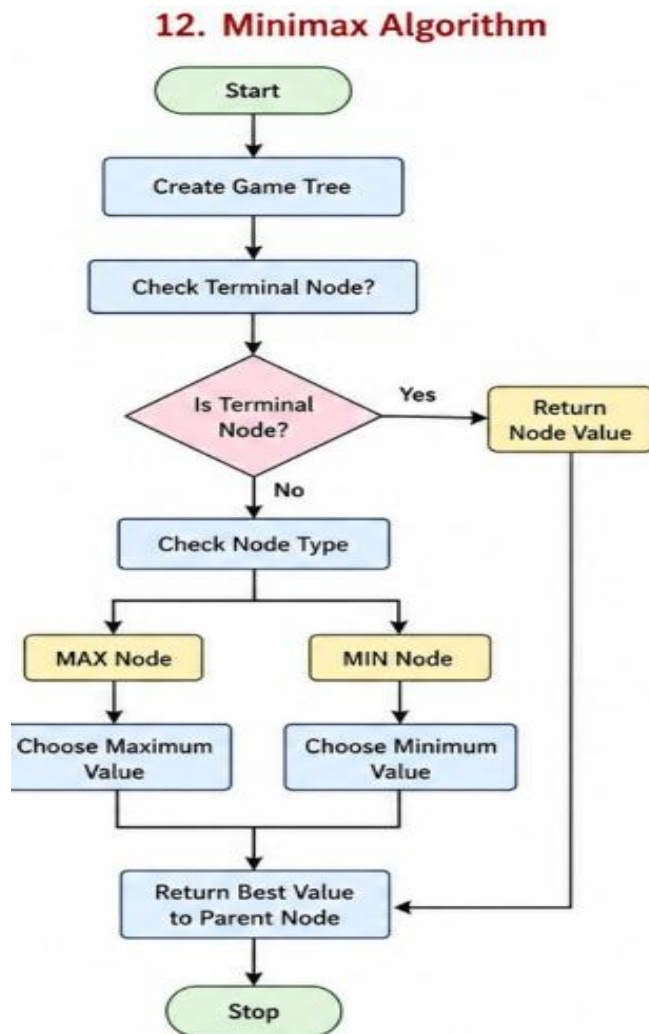
Objectives

- To understand the Minimax technique.
- To represent a game using a game tree.
- To select the best possible move.
- To apply MAX and MIN operations.

Algorithm

1. Create the game tree.
2. Check the terminal node.
3. MAX selects the maximum value.
4. MIN selects the minimum value.
5. Return the best value.

Flowchart



Code

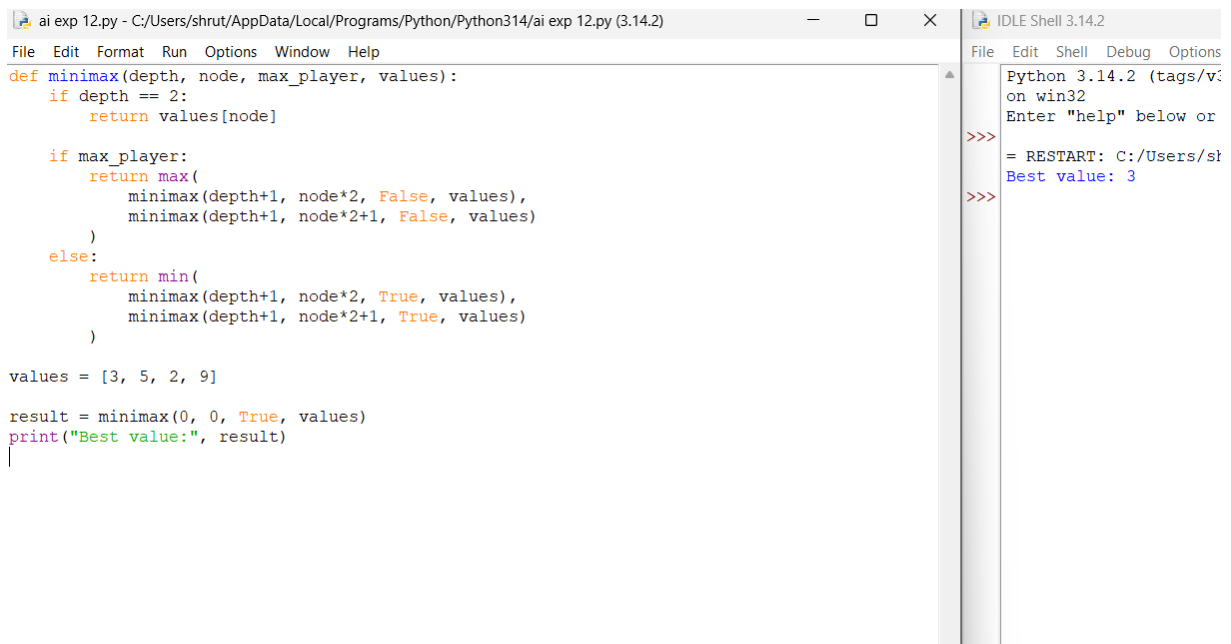
```
def minimax(depth, node,
max_player, values):

    if depth == 2:

        return values[node]
```

```
if max_player:
    return max(
        minimax(depth+1, node*2,
False, values),
        minimax(depth+1,
node*2+1, False, values)
    )
else:
    return min(
        minimax(depth+1, node*2,
True, values),
        minimax(depth+1,
node*2+1, True, values)
    )
values = [3, 5, 2, 9]
result = minimax(0, 0, True,
values)
print("Best value:", result)
```

Output



```
ai exp 12.py - C:/Users/shrut/AppData/Local/Programs/Python/Python314/ai exp 12.py (3.14.2)
File Edit Format Run Options Window Help
def minimax(depth, node, max_player, values):
    if depth == 2:
        return values[node]

    if max_player:
        return max(
            minimax(depth+1, node*2, False, values),
            minimax(depth+1, node*2+1, False, values)
        )
    else:
        return min(
            minimax(depth+1, node*2, True, values),
            minimax(depth+1, node*2+1, True, values)
        )

values = [3, 5, 2, 9]

result = minimax(0, 0, True, values)
print("Best value:", result)
|

Python 3.14.2 (tags/v3.14.2:~b20230305) on win32
Enter "help" below or

>>>
= RESTART: C:/Users/shrut/AppData/Local/Programs/Python/Python314/
Best value: 3
>>>
```

Result

The program successfully finds the best value using the Minimax algorithm.

Conclusion

Thus, the Minimax algorithm was successfully implemented for gaming using Python.